EXHIBIT 2

UNITED STATES DISTRICT COURT

EASTERN DISTRICT OF LOUISIANA

Civil Action No. 2:24-CV-228-WBV-DPC

KATHLEEN MOLLER,

Plaintiff,

vs.

MARTIAN SALES, INC.; JOPEN, LLC;
JOHNSON FOODS, LLC; LP IND., LLC;
CAG HOLDINGS, LLC; RMH HOLDINGS, INC.;
ABC INSURANCE COMPANY and John Does 1-4,

Defendants.

ORAL AND VIDEOTAPED DEPOSITION OF

RONALD WHARTON, M.D.

VIA ZOOM

Monday, February 23, 2026

10:00 a.m.

REPORTED STENOGRAPHICALLY BY:

Jennifer Miller, RMR, CRR, CCR-NJ,
CCR-WA, CCR-NM, CALIFORNIA CSR#14652
Notary Public: NJ, NY, PA, DE

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PROCEEDINGS

THE VIDEOGRAPHER: This is File Number 1 in the remote video deposition of Dr. Ronald Wharton in the matter of Kathleen Moller versus Martian Sales, Inc. et al. in the United States District Court, Eastern District of Louisiana on February 23, 2026. The video time is 10:03 a.m.

My name is Ingrid Rodriguez. I am the legal videographer from Digital Evidence Group. The Court Reporter is Jennifer Billstein-Miller in association with Digital Evidence Group.

Due to the nature of remote reporting, please pause briefly before speaking to make sure all parties are heard completely.

Would counsel please state your appearances for the record.

ATTORNEY WAGAR: My name is Chip Wagar. I'm with the firm of Wagar Hickman. I'm here with Jennifer Avallone, and I represent the plaintiff, Kathleen Moller.

ATTORNEY PAYTON: My name is Gwendolyn Payton. I represent the defendants. Along with me here today is Hillary Rightler in my office.

THE VIDEOGRAPHER: The Court Reporter will now swear in the witness.

RONALD WHARTON, MD, after having been first duly sworn, was examined and testified as follows:

- - -

EXAMINATION

- - -

BY ATTORNEY PAYTON:

Q. Good morning. As I said just a moment ago, my name is Gwendolyn Payton. It's nice to meet you.

Can you please state your full name for the record.

A. Ronald H. Wharton.

Q. And what is your business address?

A. 300 Community Drive, Manhasset, New York 11030.

Q. And are you employed?

A. Yes.

Q. And what is your employer?

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patient, and they had an untoward complication which really can't be prevented. And I gave a deposition, and I was subsequently removed from the case.

Q. What is the third matter in which you have been deposed as a defendant?

A. This was a case also at Montefiore, again, probably about five years ago -- I'm sorry. Strike that. About five years before I came to Northwell, so probably about 11 or 12 years ago as well -- of a patient who was admitted for a evaluation of a uncommon condition called constrictive pericarditis, which can be a complication of previous open-heart surgery. I wrote an admit note and suggested an evaluation, which was perfectly appropriate.

The patient had subsequent complications related to invasive procedures that were done after she was no longer in my care. And, again, I'm at a loss to explain why plaintiff counsel felt it necessary to name me. I gave a deposition.

That case has not yet been fully settled. However, I have every reason to

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anticipate that, again, I will be removed from the case.

Q. So that case is still pending?

A. Correct.

Q. What jurisdiction is it in?

A. Same. It's the Bronx.

Q. Okay. Do you know if those cases are in federal or in state court?

A. I believe they're all in state court. As I said, the others were long gone. The third, I would imagine, would be in state also.

Q. The first two cases you talked about in which you were a defendant and you gave deposition testimony, you say you were removed from the case.

Was it, in fact, that those cases were settled?

A. If they were settled or not with other parties, I can't tell. I had -- I personally had no -- I was removed as a defendant.

There was no -- there was no, quote/unquote, settlement on my part or from the malpractice carrier that covers Montefiore.

Q. So your testimony is you were

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actually dismissed from the case?

A. Correct.

Q. Okay. Are there any other malpractice cases in which you were named other than the three that you told me about?

A. No.

Q. And just to confirm, the only time you've ever testified in trial is in that one patent case you told me about?

A. Correct.

Q. Okay. Any other involvement in litigation, either as a plaintiff or a defendant or an expert witness, other than the matters you've already told me about?

A. Over the years, I've looked at a number of cases and given attorneys opinions, but those never went on to depositions or the like.

Q. Anything else?

A. Not that I can remember.

Q. Have you ever been an expert witness in a kratom-related matter?

A. No.

Q. This is the first time?

A. Yes.

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Q. Have you ever been involved as an expert witness in a matter involving 7-Hydroxy products?

A. No.

Q. Have you ever been an expert witness in a matter involving pseudoindoxyl products?

A. Which products?

Q. Pseudoindoxyl.

A. Not familiar with that, no.

Q. What did you do to get ready for the deposition today?

A. Well, I've been immersing myself in the literature that has emerged in various cardiovascular journals related to kratom. I've read reports from the website of the United States Food and Drug Administration.

Q. Anything else that you did to -- oh. Sorry.

A. And I was provided one -- I was provided with depositions and summaries of the depositions of two of their defense witnesses.

I was provided with one article that was recently published -- I'll tell you in which journal. Let me -- give me one second.

I was provided with one article

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but safety.
Q. In a drug trial with FDA?
A. Yes.
Q. Okay. Are you aware of whether anyone has attempted a drug trial related to kratom?
A. No. I'm not aware of any formal drug trial related to kratom. I just know that kratom is not approved in the United States for human consumption to treat any condition.
Q. And under what regulatory body would it be approved to treat a condition?
A. Well, I believe the FDA. The FDA is what gives drugs their indication. When -- we give medications because it has an FDA approval. Sometimes drugs are given for off-label uses, but they still have FDA approval for other drugs -- for other indications, rather.
I -- I am not aware that kratom has any studies that give it an approval to treat any specific condition. And if I'm wrong, please let me know.
Q. Are you under the assumption that anybody in this case is arguing that kratom has

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been approved as a drug?
A. I thought that's what you were asking me. If I'm wrong, my apologies.
Q. Do you believe that your role here today is a scientist or an advocate?
A. More a scientist. I was asked to review the case of a young woman who died suddenly. I've given opinions as to what I think, like, happened under most probable conditions. The most likely cause. That's all I've done.
Q. And you're a cardiologist, right?
A. Yes.
Q. So is your opinion here today limited to the area of cardiology?
A. A young woman died. I looked at the case. I looked at the -- I reviewed the findings of the autopsy, both the macroscopic, the microscopic, the toxicology, and I rendered an opinion based on all probability in my -- based on my background and experience as a cardiologist.
Q. So your opinion is based on your expertise in cardiology.
Is that fair?

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A. Yes.
Q. Does it go beyond cardiology?
A. All cardiologists have gone through internal medicine training first, so we all have that background. But I do believe that the death in this case was a cardiovascular event, specifically an arrhythmic event, which I've already opined on.
Q. I understand. I'm just asking a simple question.
When the Court asks you "What's your expertise in?" I'm unclear what you're claiming.
A. I'm a cardiologist. No more, no less.
Q. Okay. And you're not here to give expert opinions in areas beyond cardiology, right?
A. No, but cardiology is a fairly large area. It does include any relevant -- cardiology includes cardiovascular pharmacology, and it includes any knowledge of the potential for drugs to cause any harm to the heart in any way or form.
Q. So let's go back to the issue of the

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articles that you were given by the plaintiff's lawyers in this case.
They brought your attention to the Huestis 2026 study, which you hadn't seen before, correct?
A. Correct.
Q. Anything else they gave you?
A. Everything else that I reviewed was my own literature search, and I believe those articles were exchanged.
Q. Okay. And you said you were -- you got some deposition transcripts.
Is that fair?
A. Yes.
Q. Okay. And you also got some summaries of those transcripts.
Did I understand that right?
A. Which were much easier to read than the actual depositions themselves.
ATTORNEY PAYTON: Okay. So, Chip, I'll do a request for production for the production of the deposition summaries that were provided to Dr. Wharton in advance of this deposition.

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for identification.)

- - -

BY ATTORNEY PAYTON:

Q. Have you ever seen this document before?

We can scroll so you can see the totality of it.

A. This is -- I didn't -- I provided these articles to counsel. I didn't see the way they formatted it and sent it to you. But these are things that I believe I've reviewed at some point along this journey.

Q. When did you provide these articles to plaintiff's counsel?

A. At various times as I came across them.

Q. So not all at once?

A. I may have last week touched base with Mr. Wagar to ensure that they had all of the things that I had seen.

There was one, the article by Cui, C-U-I, which I believe I provided. That was the last one that I provided. That would have been sometime last week. The others I think I've been providing all along.

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Q. You see this article there? [As read]:

Brown and Chan, "Elevated 7-Hydroxymitragynine Levels Found in Products Misbranded as Kratom"?

A. Yes. I saw that.

Q. And what was your understanding of the conclusion of the Brown article?

A. I don't remember it.

Q. Okay.

A. I'd have to look at it again.

Q. And what is your understanding of products misbranded as kratom, if any?

A. I'm not sure I can answer that question the way it's phrased.

Can you list -- can you repeat the question, please?

Q. Sure. Let me ask a different question that might be better.

What is the level of 7-hydroxymitragynine in the product that is alleged that Harmony Moller consumed?

A. You mean what was her level at autopsy?

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Q. You can answer that question and then I'll go back to my main question.

What was the level of 7-hydroxymitragynine found in the autopsy of Harmony Moller?

A. I'd had to look. I'd have to look at that. I can dig it up in the autopsy, which I have on my computer somewhere. If you have it in front of you, I would believe it. I wouldn't dispute it.

Q. Okay. Would you agree with me that -- or would you not dispute that the level of 7-hydroxymitragynine is not found in the autopsy?

A. No. I believe it was found.

Q. Okay.

A. I'd have to look, but I do believe there was some there.

Q. And is that important to your opinion?

A. Yes.

Q. In what way?

A. The case does not center around the fact that it was implicated in her death. If it wasn't there, then we wouldn't have a case,

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would we?

Q. 7-hydroxymitragynine found on the autopsy is central to the case is your opinion?

A. Mitragynine is. 7-hydroxymitragynine is a metabolite, and both of those substances are implicated in causing ventricular arrhythmias.

Q. So the presence of 7-hydroxymitragynine is important to your opinion?

A. Both mitragynine and 7-hydroxymitragynine, yes.

Q. And tell me, what is the difference between mitragynine and 7-hydroxymitragynine?

A. So the 7-hydroxy is a metabolite of the parent compound. It just means that the seventh carbon in the organic molecule has been hydroxylated with a hydroxy group. That's an OH group.

Q. Is there 7-hydroxymitragynine present in the actual product itself?

A. I've never studied kratom formally. I know that kratom has mitragynine, or however you pronounce it.

And I know that the liver

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metabolizes it to the 7-hydroxy metabolite.
Q. How do you pronounce it?
A. I thought it was pronounced mitragynine. If it's pronounced another way, then I defer to you.
Q. So, again, my question was a little different, though.
Is 7-hydroxymitragynine found in the actual product itself? Do you know?
A. I don't know whether it is or if it comes as a metabolite. I think it's a metabolite, but I could be wrong.
Q. You're not sure?
A. I believe it's a metabolite. That's my understanding.
Q. So only produced in the human body?
A. That's my understanding. If I'm wrong, it doesn't affect my opinion, but that's my understanding.
Q. So tell me, what is pseudoindoxyl?
A. I don't know.
Q. Have you read any literature about pseudoindoxyl?
A. No.
Q. Have you done any research on kratom?

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A. No.
Q. Have you published any papers on kratom?
A. No.
Q. Were you familiar with kratom prior to receiving this case?
A. No.
Q. Have you ever treated any patient for kratom exposure?
A. No.
Q. In your clinical practice, have you ever encountered kratom?
A. Fortunately, no.
Q. So back to Exhibit Number 1.
Just so I'm summing this up, these are the things you reviewed, and you provided to plaintiff's counsel a list of what you reviewed?
A. These are things that I have read during the course of my endeavors in this regard. And as I've read them, I have shared them with counsel.
Q. What is your role as the director of the hypertrophic cardiomyopathy program? What do you do?

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A. So hypertrophic cardiomyopathy is the most common genetic cardiomyopathy. It is very prevalent, with an estimation of somewhere between 1 in 250 to 1 in 500 people. It's probably in the vicinity of 1 in 300.
It's a very complex condition that requires a deeper understanding than, I would say, the average cardiologist. Many of us subspecialize. This is one of those sub-subspecialties, if you will.
I treat a very, very large practice of patients with this condition, and I coordinate their care with other subspecialists within my hospital system because they require many subspecialists.
I'm also very active in clinical trials, both Stage II and Stage III, for emerging molecules that are coming out right now to perhaps improve the quality of care for these patients in the future.
Q. So am I understanding that hypertrophic cardiomyopathy is fairly rare?
A. No, your understanding is inaccurate. It is fairly common.
Q. Okay. Of your practice, what

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percentage is clinical versus academic or administrative?
A. It's 100 percent clinical. I do run a program that requires signing a lot of forms for the FDA because we're involved in clinical trials and agreements with drug companies that are sponsoring trials, but I'm always a clinician. That's what I do.
I read echocardiograms. I perform transesophageal echocardiograms. I teach a lot of echocardiography because we have a very active fellowship program. But the -- I do other things as well.
I'm the director of the ultrasound school at the institution where I work training new sonographers. I coordinate a program with sleep medicine because cardiovascular disease and sleep apnea have a lot of intersection. This is all clinical work.
The hypertrophic cardiomyopathy program takes the -- a large chunk of my clinical practice because it's incredibly busy and getting busier.
Q. When you say "a large chunk," could

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Q. Have you ever presented any material related to kratom?
A. No.
ATTORNEY PAYTON: I'd like to take a short break just to have a bathroom break.
THE WITNESS: Sure.
ATTORNEY PAYTON: Okay. We'll come back, is it okay, in five minutes?
THE WITNESS: Sure.
THE VIDEOGRAPHER: The time right now is 11:12 a.m. We're off the record.
- - -
(Whereupon, a short recess was taken.)
- - -
THE VIDEOGRAPHER: The time right now is 11:19 a.m. We're back on the record.
BY ATTORNEY PAYTON:
Q. Are you holding yourself out to be a kratom expert?
A. I don't know what a kratom expert is.
Q. Okay. Well, then are you holding

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yourself out to be one?
A. I'm a cardiologist. That's all I'm holding myself out to be. We're always learning. As new substances come out and they may have effects on the heart, we're expected to know them.
Q. Is it fair to say your opinions on kratom in this case are based solely on the articles and references that you've cited in your report?
A. Correct.
Q. Do you know what specific alkaloids are present in kratom?
A. I am told -- I'm sorry. Let me rephrase that.
I have read that there are approximately 40. I don't know their names, but I am told that there are -- I'm sorry. My apologies.
I have read that there are approximately 40 alkaloids within kratom. I have read that there is more to be learned about this product than is already known from the articles that I've read.
Q. Where does kratom come from?

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A. My understanding is that it is a concentrated form of substances which are extracted from leaves in trees in Southeast Asia.
Q. When you say "concentrated form," what do you mean?
A. I don't believe -- again, this is all based on what I have read -- that substances which are marketed as kratom necessarily reflect the same concentrations as they may be found in plants.
Q. Why do you say that?
A. That's what I've read. I believe that's what I've read. My opinions can only be based on what I've read.
Q. Where did you read this thing about the concentrations being different than in the plant?
A. I don't recall specifically. I've read many articles, as you know, in the past several months. I honestly don't recall.
Q. What's your knowledge of the product at issue in this case?
A. Kratom.
Q. What does that mean?

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A. These are substances that are sold.
Q. I mean, do you have any knowledge of the specific product that allegedly Ms. Moller consumed?
A. I know that she took a product that's labeled kratom. I know there are several companies that market it. I know that it has 7-hydroxymitragynine in it. I know that it has metabolite that is active on ion channels in the heart. I know that there are -- I have read that there are many other alkaloids in those substances that have not been studied.
I have not actually purchased one or seen one or had any contact with any patient who has taken it.
Q. When you said there are several companies that market it, what are you talking about?
A. Well, I believe that -- I believe that there are several companies out there. I believe Martian Sales is only one of them. If I'm wrong, I'm wrong. But I don't believe they're the exclusive people who sell it in the United States.
Q. Who is Martian Sales?

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A. I believe the defendants in this case.
Q. So what is the role of Martian Sales in marketing it?
A. How they market it, I don't know. I believe they profit from it.
Q. And why do you say that?
A. They're selling it. I believe – you know, I can't tell you every little thing that I've read in the past several months, but I do believe that at one point I read that the profit margin on this substance nationally was several hundred million dollars annually. So someone is making money on it.
Q. What's your basis for saying that?
A. Should I go to the internet?
Q. Is this something that you read on the internet?
A. Yes.
Q. Okay. Are you going on the internet during this deposition?
A. I have not.
Q. Okay. What do you have in front of you?
A. You.

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Q. Anything else?
A. Nope.
Q. You've been picking up a lot of papers and shuffling them around, so I don't know that's true.
A. Right. Oh, yes, you do because I'm under oath, and I'm not perjuring myself, Counselor.
Q. Okay. You told me you have nothing in front of you, but, in fact –
A. I have – I have the articles – these are the articles that I reviewed yesterday that I was referring to. These are all from journals. These are all journal articles.
Q. Anything else?
A. No.
Q. Okay. When you're doing that behavior, it actually doesn't work because you have that hazy screen saver thing where we can't see your real background, so it makes you disappear.
A. Oh, I'm sorry. I didn't mean to disappear.
I am not intending in any way to

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perjure myself, Counselor. The articles that I have in front of me that I've been looking at are the articles from the Journal of American College of Cardiology.
There's one article from the FDA website and one article which is in my report from Clinical Translational Sciences from 2023 written by L-I et al. I have no other things in front of me nor have I been going on the internet during this deposition.
Q. Okay. So back to my question.
This profit margin testimony that you're giving us, what's your basis?
You just told us you're under oath. So –
A. Yes, of course. My basis is – yes, of course. So over the past several months, I've looked into this. That's a natural thing to do. You're retained on a case. It involves a substance. As I have already stated under oath, I had not been familiar with this until I was approached by the law firms. So when you learn about something, you learn about it and you read about it. And at some point, I think I read – I don't know specifically – that

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this is a very, very profitable industry.
Q. Is that part of your expert opinion here?
A. I'm being honest in what I read. If you want to call it part of my expert opinions, so be it.
Q. Okay. You are planning on so testifying at trial?
A. Between now and trial, I'm sure I will verify anything that I say to make sure that I'm accurate. But I'm being as honest as I can be.
Q. Okay. So your – you will be prepared to testify about the profit margin related to this product at trial?
A. I could research it if I needed to.
Q. And this is information that you got from the internet?
A. Yes. I would have no other mechanism for acquiring such information.
Q. So – and that's sufficient to be a basis for your expert testimony at trial?
A. It's as sufficient as I could possibly be. I don't get financial reports from companies directly that sell this.

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Q. So I want to go back to your testimony about the product at issue in this case.

You said it's kratom.

A. Uh-hum.

Q. Is that right?

A. Yes.

Q. Okay. The reason I'm cuing you is because "uh-hum" and "huh-huh" don't translate.

A. I'm sorry. Yes.

Q. Okay. Try not to talk over me.

So other than kratom, do you have any knowledge about what the product is?

A. I'm not sure I understand the question.

Q. Is it an extract?

A. I believe it's an extract from leaves from Southeast Asia. I believe it's marketed as powders, as drinks.

Q. What do you understand that Harmony Moller consumed in this case?

A. I understood that she consumed a kratom product. I do not know the specific product that she took.

Q. And you don't know what kind of

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product it was?

A. I don't recall. I believe it was a liquid, but I don't recall specifically.

Q. Is whether or not it was an extract product or not extract product important to your opinion?

A. No.

Q. And why not?

A. Because mitragynine and its metabolite have been emerging more and more in the cardiovascular literature as substances of concern because of their ability to provoke dangerous heart rhythms in susceptible individuals.

The mechanism of how it got there specifically, whether it was ingested one way or another, I don't believe is pertinent.

Q. Do you have any expertise in determining cause of death?

A. I read the pathologist's report.

Q. No. I'm asking a general question. I understand --

A. I'm not a pathologist.

Q. Do you determine cause of death as part of your clinical work?

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A. Fortunately, my patients aren't dying every day.

Q. So no?

A. "No" is not the right answer. All of us, as physicians, have patients die on us. It comes with the territory. And frequently we do know what happened. Of course, these are patients that are in our care.

Are cardiologists, as a rule, in the habit of opining on cause of death on patients that we've never seen before? I don't think anyone is besides a pathologist.

Q. So no?

A. You can read a history and you can draw reasonable conclusions. You can look at a pathologist's report and infer reasonable conclusions.

Q. My question is pretty simple.

Do you determine cause of death as part of your clinical work?

A. Only when one of my patients dies and -- sometimes.

Q. How frequently do your patients die?

A. Fortunately, not very.

Q. Do you certify causes of death?

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A. All physicians do at some point in our training. We're all asked to sign death certificates and write up a cause of death on the death certificate.

I've had to do that in my training, as do all physicians when they're training at some point.

Have I had to do that recently? Fortunately, no. It's not particularly that pleasant.

Q. Okay. So are you testifying that you certify causes of death in your current role?

A. I can't answer that question the way it's asked. If a patient of mine has a heart attack and they died from it, I can say they had a heart attack and they died from it.

Do I generally certify causes of death in patients that I've never seen before? Well, of course not. That's not what we do.

Q. When was the last time you signed a death certificate?

A. Not since I've come to North Shore.

At Montefiore, probably the last time would have been, I don't know, 15 years ago. Maybe 10 years -- maybe 12 years ago. I

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don't know. Something like that. Nothing recently.
Q. And you've never certified a cause of death to be related to kratom or mitragynine?
A. No.
Q. Are you a medical toxicologist?
A. No.
Q. Do you have any other specific training or expertise in medical toxicology?
A. No.
Q. Do you have any specific understanding of how mitragynine is assessed in a toxicology report?
A. Only as a level – as a serum concentration.
Q. How do they do that?
A. It's a blood test. It's an assay.
Q. Any other knowledge?
A. No.
Q. Do you know if they can test for 7-hydroxymitragynine?
A. Yes, they can.
Q. And you're sure of that?
A. Fairly, yes.
Q. Okay. Do you know if mitragynine is

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part of a standard panel for forensic blood tests performed at autopsy?
A. I would have no knowledge of that, but standard blood tests evolve with the rest of the world.
Q. What do you see now?
A. As of now, I don't know – I cannot – I can neither confirm nor deny.
Q. Okay. Do you know if it needs to be – if mitragynine needs to be specifically requested?
A. I would not know that either way, and there may be considerable variability from state to state and from county to county and from jurisdiction to jurisdiction. I don't know the answer to these questions.
Q. Okay. Do you know whether other drugs must be specifically requested as part of a standard forensic blood test performed at autopsy?
A. I'm not a pathologist.
Q. What is a safe level of mitragynine?
A. I don't know that there is one.
Q. Do you know one way or the other?
A. It hasn't been studied formally to

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say what's safe. And what is safe? Humans are not a homogeneous species. What is safe in one person may not be safe in another. We know that with many, many drugs for which we are required to have knowledge.
Q. What was the dose level that had no adverse effects in the FDA study from 2025? Do you know?
A. Of course I don't. I told you I didn't remember the study.
Q. Okay. Do you know if there's a lethal dose of mitragynine?
A. I don't believe anyone knows that because humans are not all alike. What may be okay for one person may not be okay for another, which is very, very true for medications which have actually been formally tested and adjudicated by the FDA.
Q. So, no, you don't know if there is a lethal dose of mitragynine?
A. I don't believe anyone knows that.
Q. What is the characteristics of postmortem redistribution with mitragynine?
A. I don't know the answer to that question.

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Q. So we have your report listed as Exhibit Number 1.
Do you have a copy of it in front of you?
A. I do not.
Q. On page 1 – I can put it up. Just say the word if you want it in front of you because I didn't –
A. Sure. I don't have it in front of me.
Q. Okay. Most of the questions that I have just move faster. I think we'll be noncontroversial about what it says. But if for some reason I'm asking you a question and it feels like "I want to look at it," just say it and we'll put it up on the screen.
Is that fair?
A. Yes.
Q. Okay. On page 1 of your report, you have a list of what you reviewed, and it's in some bullet points.
Do you recall that?
A. That's something that I would do. That would be my custom and practice.
Q. And there's about six things that you

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reviewed related to the case?
A. Uh-hum.
Q. Is that a "yes"?
A. I believe you.
Q. Yeah.
A. Like I said, I don't have it in front of me, but I have no reason --
Q. Yeah, let's put it up. Because I want to make sure that I'm not misleading you.
I'm referring specifically to this part at the bottom of page 1 of Exhibit 1.
A. Okay.
Q. And so these are the things it says that you looked at in preparing your report; is that correct?
A. Yes.
Q. Is there anything else that's not listed here that you looked at?
A. I don't remember. This is from November. I don't remember -- if this is what I wrote, then this is what I wrote.
Q. Okay. And we can trust that it's true.
Is that fair?
A. I would not write anything otherwise.

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Q. You state in your report that kratom packages, and I quote, [as read]:
"were found in the bedroom of the descendent at the time of her death."
Do you recall that?
A. I typed that; therefore, I must have seen it. It was probably provided to me, yes.
Q. Okay. Because I was going to ask you: What's your basis for saying that the kratom was found in the bedroom?
A. It would have had to come from counsel. It would have to have been provided to me months and months ago when I was initially contacted about this case.
I would have no other way of knowing that.
Q. Did you review any of the crime scene materials?
A. I reviewed only what was sent to me. If there were images that were sent to me, I reviewed them. I wasn't there. So I was only -- you know, if photographs were sent to me of things that were there, then that's what I looked at.

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Q. Well, they're not listed in your bullet points here, so I can assume that you didn't look at them?
A. Correct. If I didn't see anything, then -- I wouldn't write anything in that report unless it were accurate.
Q. Okay. I noticed that -- well, were you aware that there were over 150 crime scene photos taken in this case?
A. No.
Q. Would they be something that you want to see?
A. I would leave that to the discretion of plaintiff's counsel if they thought it was pertinent.
Q. Was there anything that you wanted to see that you asked for that you didn't get?
A. No.
Q. Did you review the police body camera footage?
A. Body camera footage? No, I have not seen any body camera footage --
Q. Okay.
A. -- that I know of. I mean -- no. What I typed is what I saw, because my custom

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and practice is to be transparent and honest. If it's not there, I didn't see it, and that's it.
Q. So you did not see the fire department/EMS report, correct?
A. Not that I recall, no.
Q. Do you know whether there were other supplements found at the scene?
A. Supplements? No. Not that I can recall.
Q. Would that be important to you?
A. Not necessarily.
Q. Okay. Let's take a look at 8B.
A. Are we waiting for something?
Q. Yes.
A. Oh, okay.
Q. Have you seen this image?
A. I don't remember seeing this image.
Q. Okay. And you see --
A. I may or may not have, but I don't remember it.
Q. But you'll agree with me it's not listed on your --
A. I don't remember seeing this.
Q. Okay. I'm sorry. Let me finish the

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question.
You'll agree with me it is not listed on the bullet points of what you reviewed?
A. Correct.
Q. Okay. And you see in there, there is an Apetropics One Chews CBD product there on the lower left?
A. Yeah. I see it.
Q. Okay. That's a CBD, Lion's Mane and Cordyceps product?
A. Yes.
Q. Okay. And there's a Natrol, N-A-T-R-O-L, high-caffeine energy supplement, correct?
A. Yes.
Q. Some alcohol?
A. Yeah. I don't know what that is, but –
Q. I think it's bourbon.
A. Okay.
Q. Were you aware that multiple empty Jameson whiskey bottles were found at the scene?
A. I have no knowledge of that.

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Q. Okay. And you see there's also a Zantrex Black, an energy weight loss supplement.
Do you see that there?
A. Which one is that? I don't know where you are. I see alcohol. I see Jameson something.
ATTORNEY PAYTON: Okay. Can we go back to 8B, please.
- - -
(Whereupon, Exhibit 3 was marked for identification.)
- - -
BY ATTORNEY PAYTON:
Q. Oh. See where it is in the upper right, in that bag?
A. Oh, all the way up there?
Q. Yeah.
A. It's upside down.
Q. It is.
A. What is that?
Q. It is a energy weight loss supplement.
A. Okay. I don't know what it is. Zantrex, it looks like.

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Q. Correct?
A. I've never heard of it.
Q. Did you consider the potential impact of any of these supplements in coming to the conclusion in your report?
A. I didn't know that these substances were there. However, it doesn't alter my opinion.
Q. Do you know whether the – any of the toxicology lab tests would have detected these substances?
A. I don't recall – Zantrex isn't a substance. It's a – probably a – I am assuming it's probably a name of something that is marketed. I don't know what's actually in there.
Q. Okay. So you don't know if they tested for it or not?
A. I can't confirm or deny.
Q. And is it possible that any of the supplements or substances or products that we see here in Exhibit Number 3 could have had an effect on prolonging the QT interval?
A. Not of which I'm aware. Certainly none of them have made their way into the

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cardiology journals the same way kratom has.
Q. Have you done research on them?
A. A cardiologist – no. On those particular products? I don't even know what's in those products. However, we all try to stay current on literature. I'm not aware of caffeine, Naprosyn, CVS Daytime Cold and Flu causing a sudden death. And I have no idea – as a rule, alcohol in and of itself does not cause sudden death.
Q. I'm not asking about sudden death. I asked you about prolongation of the QT interval.
A. Right. So I'm confident that CVS does not market a cold and flu product that prolongs the QT interval. Alcohol does not prolong the QT interval. Caffeine does not prolong the QT interval. Naprosyn does not prolong the QT interval.
I don't know what is in One Chews or Zantrex. So I can't confirm or deny whether they don't.
However – I can't confirm whether they do or don't.
Q. Okay. Are you familiar with

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methylphenidate?
A. I've come across it. It's a stimulant.
Q. Okay. Are you aware that Ms. Moller had an empty prescription pill belonging to another individual for methylphenidate?
A. Could you say that again, please?
Q. Yes.
Are you aware that Ms. Moller had an empty prescription pill bottle by her body belonging to another individual for methylphenidate?
A. No, I was not aware of that. I am not aware that methylphenidate has any QT prolongation off the top of my head. I'd have to look it up, but I don't recall that it does.
ATTORNEY PAYTON: I'm marking that as Exhibit 4.
- - -
(Whereupon, Exhibit 4 was marked for identification.)
- - -
BY ATTORNEY PAYTON:
Q. Have you seen this image before?
A. No.

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Q. Is it something that you would want to investigate in your opinion?
A. Having been a cardiologist for as long as I have, I think if methylphenidate were a QT prolonger, it would have been on my radar. I do not recall that it is.
Q. But you don't know for sure?
A. I'm pretty sure it's not. I mean, I could look it up, if you'd like me to right now.
Q. Do you know whether any of the toxicology testing was able to test for methylphenidate?
A. I would have to look again. I did review the toxicology. I don't have it right in front of me, so I don't recall whether it was there or not. However, I do not believe that methylphenidate is proarrhythmic.
Q. Okay. I know you want to tell me you don't believe it is. I'm just asking you: Did you consider it or not?
A. I didn't consider it because I didn't know that she had it.
Q. Okay. So are you familiar with – I always say this wrong – Ashwagandha? It's

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spelled A-S-H-W-A-G-A-N-D-H-A.
A. No.
Q. Do you know whether Ms. Moller was found with a bottle of Ashwagandha?
A. I would have no knowledge of that. I'm not familiar with it.
Q. Did you know that also in her presence were drug paraphernalia?
A. Which paraphernalia would that be?
Q. Let's take a look here. Let's take a look at 8F.
As we're loading up Exhibit Number 5, you'll see glass pipe, grinder, numerous other supplements.
- - -
(Whereupon, Exhibit 5 was marked for identification.)
- - -
ATTORNEY PAYTON: We'll mark this as Exhibit Number 5.
BY ATTORNEY PAYTON:
Q. Have you ever seen this image before?
A. No.
Q. We have here what's been marked as Number C.

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Have you ever – or 6. Sorry.
Have you ever seen this image before?
- - -
(Whereupon, Exhibit 6 was marked for identification.)
- - -
THE WITNESS: No. I have not.
BY ATTORNEY PAYTON:
Q. Okay. And you'll agree with me there is drug paraphernalia and multiple prescription bottles?
A. There are multiple prescription bottles. You'll have to forgive my naivete. I don't know what drug paraphernalia looks like.
Q. Would you trust me if I would say the stuff on the plate there, the foil, container, the grinder, is sometimes associated with drug paraphernalia?
A. I would have no reason not to believe you, Counselor.
Q. And then we have on the left there, there's a vape pen.
Do you see that?
A. A what?

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Q. A vape pen.
A. I trust you.
Q. We have another cup there, some green material, correct?
A. You want to put your arrow on that?
Q. Yeah. This is actually the one -- yeah. You see that big cup?
A. Yes. Is that the kratom?
Q. No. But do you know what it is? I should ask you.
A. No.
Q. Okay. And then if you move down, you'll see the pipe.
A. Uh-hum.
Q. Is that a "yes"?
A. I see it.
Q. Yeah. Let's just quickly do H, I and K.

- - -

(Whereupon, Exhibit 7 was marked for identification.)

- - -

BY ATTORNEY PAYTON:
Q. We have what's marked as Exhibit Number 7.

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Have you ever seen this image before?
A. I have not.
Q. You'll see there's numerous vape pens?
A. I take your word for it.
Q. We're showing you what's has been marked as Exhibit 8 --

- - -

(Whereupon, Exhibit 8 was marked for identification.)

- - -

BY ATTORNEY PAYTON:
Q. -- which is also a crime scene photo.
Have you ever seen this image before?
A. No.
Q. You'll agree with me it shows some supplements and numerous prescription drug bottles?
A. I see supplements. I don't know what the Walgreens or -- it looks like "Walmart" upside down.
I don't know what's in those. I can't see the labels.

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Q. Yeah. But they look like prescription bottles?
A. Yes, they do.
Q. Wouldn't you want to know what's in those?
A. I'm happy to be told.
Q. Okay. But in forming an opinion about the cause of death, wouldn't you want to know what those are?
A. Yes and no. I can give you -- would you like me to elaborate on that?
Q. I'm just -- I'm stunned that as a cardiologist you wouldn't want to know what prescriptions, drugs were involved.
A. Well, of course I would. But here's the problem, Counselor --
Q. You cannot conceivably go in front of a jury and say it doesn't matter to you.
A. Oh, well, it matters, but at the same time, it matters for multiple reasons. Yes, you want to know what the patient is taking, but at the same time, if the patient were taking something that could potentiate the effects of something like kratom, the patient wouldn't know that that interaction exists

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because there's no disclosure on the part of kratom or its products of which I'm aware that says if you're taking A, B, C, D, E, or anything else, that you shouldn't be taking our product.
So, for example --
Q. That's not my question. It's really simple, and either you can -- you're willing to answer it without disassembling or not.
As a cardiologist, would you not want to know which prescriptions this person was taking?
A. Yes, of course.
Q. And you didn't know, did you?
A. I don't -- I believe one of the bottles that you showed me was a prescription that was for somebody else's -- under somebody else's name.
I don't recall -- again, my report was several months ago, but I don't recall her being prescribed anything, and I certainly don't remember her being prescribed anything that would be proarrhythmic.
Q. Would you be alarmed to find that this person was taking prescriptions that were

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not prescribed to her?
A. I would be alarmed if anybody was taking a prescription that was not prescribed to that person. It would be helpful to know what that was specifically.
Q. Okay. But you can sit here and tell me that none of this matters?
A. I didn't say that, Counselor. Don't put words in my mouth.
Q. No. I'm just repeating --
A. I don't --
Q. I'm sorry. I'm asking a question. You wait.
A. It matters what they're taking --
Q. You need to wait for a question. I understand you're distressed, but you have to wait for questions and answer questions. That's how a deposition works.
A. Yes, Counselor.
Q. Okay.
A. I do recall her taking Phenergan --
Q. There's no question pending.
A. Sorry.
Q. I understand this is distressing to you, but go ahead and tell me what you want to

Page 91

say. I'll get it out.
A. I recall that she was -- that Phenergan was found in her system. Phenergan is prescription. I recall that Benadryl was in her system. Benadryl is over the counter.
I recall that -- I believe it's 9-THC was found in her system. I do not recall any other prescriptions that were put under her name at all. And certainly I was -- I never saw any prescriptions under her name for substances that would be proarrhythmic.
Q. I've lost the exhibit.
Was Phenergan prescribed to her?
A. Phenergan was found in her system. Whether it was prescribed to her or somebody else, I don't know.
Q. Do you know if the Zoloft was prescribed to her?
A. I do not know.
Q. Do you know if Compazine was prescribed to her?
A. I don't recall if it was prescribed to her or not.
Q. You don't know one way or the other?
A. You already showed me one bottle

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where she was taking something that was prescribed to somebody else.
So I don't know what she was taking. I know what substances were -- or at least at one time, when I looked at her toxicology report, I know what substances were found there.
Q. Okay.
A. But whether or not they were prescribed to her or she acquired them through an illicit means, I don't know. I haven't seen the bottles.
Q. With the toxicology report, do you know what they screened for and what they didn't?
A. I would assume they did a fairly extensive screen for anything -- any substances that could possibly be in her system. What they specifically screen for, you know, in Mississippi, I don't know.
Q. Okay. And this was in Mississippi?
A. I believe so.
Q. But you're not aware. You're just assuming what was screened for?
A. Correct.

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Q. I'm showing you what has been marked as Exhibit Number 9.
- - -
(Whereupon, Exhibit 9 was marked for identification.)
- - -
BY ATTORNEY PAYTON:
Q. Are you familiar with this image?
A. No.
Q. Are you familiar with the nuERA product?
A. No.
Q. Do you know what it is?
A. No.
Q. Would that be important to you as a cardiologist to find out?
A. Yes.
- - -
(Whereupon, Exhibit 10 was marked for identification.)
- - -
BY ATTORNEY PAYTON:
Q. We've marked what has been shown -- what has been marked as Exhibit Number 10. And this is actually a still image taken from the

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police camera.
And you see that product Mini Thin?
A. Yes, I see it.
Q. Do you know what that is?
A. No.
Q. Have you seen this image before?
A. Never.
Q. As a cardiologist, would you want to investigate what that product was?
A. Yes.
Q. Are you familiar with yohimbine? Y-
A. Yohimb- -- I'm sorry?
Q. Yeah, let me spell it for the Court Reporter. Y-O-H-I-M-B-E.
A. It's Y-O-H-I-M-B-I-N-E; is that correct?
Q. I don't believe so, but it doesn't matter.
Are you familiar with yohimbine?
A. Remotely from my days in medical school pharmacology. I believe it's an alpha agonist. It's been a long time, decades since I looked at it.
Q. As a cardiologist, would you want to

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investigate the possible effect of yohimbine on this death?
A. My recollection is that yohimbine does not have a proarrhythmic effect. I believe it's an alpha receptor agonist or blocker. We're going back many years in pharmacology, but it is not on my radar as a substance that has effects on the QT interval or proarrhythmia in general.
Q. Well, does oxycodone have an effect on the QT interval?
A. No.
Q. But it can kill you, right?
A. Yes.
Q. Are you aware that a prescription bottle of oxycodone was also found at the scene?
A. I was not aware of that. I do not recall any narcotics and narcotic concentrations in the toxicology report.
Q. Okay. But agree with me there are more ways to die than just a prolonged QT interval and a heart defect, correct?
A. Of course.
Q. You reviewed the death certificate,

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correct?
A. Yes.
Q. And you also reviewed the coroner's report consisting of four pages; is that right?
A. Yes.
Q. Were you ever provided more comprehensive records from the coroner's office, more than just the four-page summary?
A. I think I just got the four-page summary.
Q. Okay. Do you recall reading under "Medical History" that she suffered from spina bifida?
A. I don't remember that specifically.
Q. I want to correct something I said.
The prescription that was found was hydrocodone.
Are you aware of that?
A. I am now.
Q. Okay. I may have said another opioid. I can't remember what I said. It was hydrocodone.
But you didn't know that before now?
Is that a "no"?

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A. That's a "no."
Q. Okay. Did you review the litigation support package from NMS Labs from the toxicology report?
A. The what?
Q. The litigation support package, the long-form toxicology report.
A. I looked at a toxicology report.
Q. Okay.
A. I don't -- if there were more than one, I'm not aware of it.
Q. So the full package is over 1,000 pages.
A. I can guarantee you I did not read 1,000 pages.
Q. And there are various panels that they can pull on in NMS Labs.
Are you aware of that?
A. I trust you. I wasn't aware of it.
Q. There's a basic panel, there's an expanded panel, and then there's a total tox panel.
Are you familiar with that?
A. Okay. No.
Q. Okay. Do you know which one was

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pulled here?
A. I remember a form that couldn't have been more than one or two pages.
Q. Okay. Do you know how old the blood sample was at the time they did the toxicology report?
A. No. I believe -- again, I would assume it would have been fairly quick, within a day or two, but I have no specific knowledge of the time between when the blood was corrected -- collected and when the sample was run, no.
Q. So you don't know?
A. No.
Q. Would it surprise you if it was a multiday delay?
A. I have no basis to be surprised or not. I don't know what the custom and practice is in pathology labs. And I'm sure there's considerable variability from state to state. It's a big country.
Q. And you agree with me that blood decays as it sits?
A. I'm not sure how you define blood decaying.

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Q. Okay. Is the accuracy of a toxicology pull affected by the length of time that the blood has been sitting?
A. For electrolytes, absolutely. For exogenous substances, concentrations might change, but if they're there, they're there.
Q. Are you sure of that?
A. You can't find something that wasn't there. Perhaps something could have been there that could have been degraded maybe, but I don't believe so, no. You would find it. You might not find it in the same concentration that it was there initially.
Q. Are you aware that --
A. No. I don't think you would completely miss it over the course of a few days, no.
Q. Are you familiar with the half-life of certain benzos?
A. Some benzodiazepines last longer than others.
Q. And some of them are very short, correct?
A. Correct.
Q. Are you aware of the half-life of

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7-hydroxymitragynine?
A. I would have to look it up.
Q. So it is possible that if you don't get to it fast enough, you can miss certain substances?
A. I'm not a toxicologist. We've established that.
Q. You don't know?
A. I don't think you would miss it. You might not get it in the right same concentration. Let's put it this way: You're not going to find something that really wasn't there.
Q. I'm not asking if you can find something that wasn't there.
I'm asking if you wait too long, some psychoactive substances will degrade in the bud and not be perceptible?
A. That's not part of my -- that's not part of my training.
Q. So in other words, you just don't know?
A. No.
Q. Okay. So let's look at what you actually did review.

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You did review the UCSF lab report, correct?
A. I reviewed a -- CSF -- I don't recall the acronym, but I reviewed a lab report.
Q. Okay. One or more than one?
A. I reviewed a gross anatomical pathological report, a microscopic pathological report, and a toxicology, which was more than a couple of pages, I believe.
Q. Okay. One toxicology report total?
A. I wasn't aware that there were more.
Q. I think you just said that.
But you reviewed the heart pathology report, correct?
A. Oh, yes, of course.
Q. And did you review the organ donation records from the Louisiana Organ Procurement Agency?
A. Yes.
Q. Did you listen to any of the audio recordings of the mother discussing the medical history and other family members' medical history?
A. None, no.
Q. Okay. Are you aware of whether or

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not this family has any additional medical history related to cardiac events?
A. I am not aware that there is any such history, no.
Q. Are you aware that there is not such history?
A. I do not recall that there is any history of that nature.
Q. How did the brother die?
A. I don't remember.
Q. Are you aware that he died at age 36 unexpectedly?
A. No.
Q. Do you know his cause of death?
A. No.
Q. Do you know the official cause of death for the father?
A. No. I believe he was elderly when he died.
Q. But you don't know anything about the brother?
A. I don't remember the brother. I remember that the father was -- the father did not die prematurely.
Q. Are you aware that the other brother,

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the one living child, has recently had cardiac tests done?
A. No.
Q. Are you aware of the results of those tests?
A. No.
Q. Would your opinions be affected if the brother had had premature sudden cardiac death?
A. It would depend on what caused it.
Q. But it's certainly something, as a cardiologist, you would want to take into account, correct?
A. There are multiple causes of sudden cardiac death.
Q. No. That's not my question. Listen to my question.
A. Yes, I would want to know.
Q. Of course.
And you haven't been provided that information, correct?
A. No, I have not.
Q. Were you provided Harmony Moller's dental records?
A. No.

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Q. Or any records related to a cyst that she had removed?
A. No.
Q. Are you aware of whether she was on antibiotics?
A. I do not recall any antibiotics, no.
Q. Do you know if she was experiencing a dental infection?
A. No.
Q. Do you know if she was on birth control?
A. No.
Q. But you understand that Ms. Moller had been living in Asia for 12 years, somewhat close to the time that she died?
A. Yes. I was told that.
Q. Okay. Did you ask for her recent medical records?
A. I was provided everything that counsel thought was pertinent to provide me.
Q. Did you ask for anything?
A. I asked for anything that they thought was relevant.
Q. But you're the doctor, right?
A. Yes, I'm the doctor.

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Q. Okay. So when I go to the doctor, I don't just tell the doctor the things I think are pertinent. I ask the doctor -- I rely on the doctor to tell me what they need to know to do their job, right?
A. Yes. I'm being retained by plaintiff's counsel in a case. I would -- I think it is a fair statement to say that plaintiff's counsel would send me anything that they would consider relevant medically.
Q. And you rely on their determination of what is relevant?
A. Yeah. I think that's a pretty good thing to do. I don't think that they would exclude things just for the hell of it.
Q. Okay. Do you think it's usual to have no medical records for two years?
A. When somebody has been in Asia for 12, yeah, it's very possible.
Q. How long had she been in the United States before?
A. I don't know. But young people don't make it a habit of necessarily going to doctors on a regular basis. There are a lot of young people out there that have no medical records

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simply because they feel fine and they don't go to doctors.
Q. But did you even ask, "Hey, are there any medical records for this person?"
A. I asked for anything that they thought was relevant.
Q. Okay. We know that --
A. It's not for me -- I don't know what's out there. They know that.
Q. Okay. You sound kind of defensive about this. But, in retrospect, would it have been wise to ask for medical records before you rendered a cardiac opinion?
A. I don't recall what I asked for and what I didn't ask for.
Q. Okay. But you'll agree with me medical records matter, right?
A. If they exist.
Q. Okay. But -- okay. So you have somebody who does have a congenital heart issue, right?
A. How did you arrive at that conclusion?
Q. Well, we know that her heart is abnormal. Agree?

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A. We know that her heart is abnormal. We have no idea whether that was congenital or acquired.
Q. Okay. So let's just stick with the abnormal heart.
We don't know if it's acquired or if it's -- if she's born with it, right?
A. Correct. We don't know.
Q. Wouldn't you want pediatric records from this person to understand what's going on with the heart?
A. Most likely, if there were pediatric records that showed that she had a congenital heart abnormality, that would have already been disclosed.
Furthermore --
Q. That's not my question.
Is this how you do your practice? That you just say, well, you know, maybe they'll give it to me, maybe -- I mean, don't you inquire into what's going on? I want to do a fulsome analysis so that I can testify within a degree of reasonable medical certainty?
A. Counselor, the question is offensive.

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I know how to do my medical practice. I do a damn good job of it.
This is a medical malpractice case. I have to assume that counsel provides me everything they consider that's pertinent.
Furthermore --
Q. No, I know you're --
A. Furthermore -- I'm entitled to finish -- don't talk over me.
Q. No one is talking over you. Wait for a question.
A. No, no. Furthermore --
Q. Wait for a pending question.
The question is: Did you think that pediatric records would be important to rendering a cardiac opinion?
That was the question that's pending.
ATTORNEY WAGAR: No, I object. You talked over him. You didn't let him finish. I object to the form of the question. He's entitled to finish his answer and he wasn't finished and you talked over him. So I object.
ATTORNEY PAYTON: Okay. Fair

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enough.
BY ATTORNEY PAYTON:
Q. Okay. Go ahead and finish what you want to say, but then let's get back to actual questions.
A. Okay. There was nothing provided to me that suggested that she had any congenital heart abnormality.
Furthermore, pathologists -- let's put it this way: All doctors, I think, strive to be accurate in our descriptions. It protects our patients, and it protects us from having to deal with the likes of people of you. We don't like to be defendants.
Q. Well --
A. Pathologists --
ATTORNEY WAGAR: You're talking over him.
ATTORNEY PAYTON: Sorry.
THE WITNESS: Pathologists are not in the habit of playing the game, quote, Guess What I'm Thinking, unquote.
They have a pathology report. If they said -- if they thought that there was a congenital heart condition

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transcript from the mother?
A. I have not seen the mother's deposition.
Q. Do you have any information to -- related to Ms. Harmony Moller's history or pattern of kratom use?
A. No.
Q. And you don't know what kind of products she consumed; is that correct?
A. What kinds of kratom? I know she took kratom. I don't know what kinds of kratom she may have taken.
Q. Do you know the amount?
A. No.
Q. Do you know the reading in her toxicology report related to mitragynine?
A. I saw it. I don't recall it.
Q. Okay. Let me represent to you it's hovering somewhere about 150 nanograms per milliliter in the blood.
A. Okay.
Q. So in your understanding, is that a high or a low amount of kratom?
Of mitragynine. Sorry.
Let me ask the question again.

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I bungled it.
Is it your understanding that it's a high or a low amount of mitragynine?
A. My recollection is that it's not high.
Q. Do you know if Ms. Moller vaped?
A. I don't know.
Q. Would that be important to you?
A. I would want to know what she was vaping.
Q. And why?
A. Because if someone is taking a substance, you would want to know what the substance is because that's the only way you could determine what its potential for adverse effects are.
Q. Same question with -- related to alcohol use.
Do you have any knowledge about the frequency or existence of her alcohol use?
A. Not -- nothing other than the photograph that you showed earlier.
Q. Would that be important to you?
A. From the perspective of proarrhythmia, not per se.

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Q. What about otherwise?
A. Yes. It would be pertinent.
Q. Did you inquire about that?
A. Not that I specifically recall.
I have no knowledge that she was or was not a heavy drinker, nor I wouldn't even know how you would define that.
Q. Are you aware of the evidence that Ms. Moller may have consumed a whole bottle of Tums?
A. Tums?
Q. Tums, like the anti --
A. Yeah, sure. I don't know that she took Tums. And if I did, I wouldn't certainly know how much she took over what period of time.
Q. But are you aware of testimony from the mother on the police camera that she took an entire bottle of Tums?
A. No.
Q. Would that have any effect on your opinion?
A. I don't recall that she had any electrolyte disturbances that would have reflected toxicity from calcium carbonate,

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which is what Tums is.
Q. Is there toxicity from calcium carbonate?
A. There's toxicity from calcium.
Q. And what is that?
A. Well, if you're hypercalcemic, you can have -- you can have -- you can have problems.
Q. What kind of problems?
A. Tetany, you know, muscle contraction, things of that nature. It is not per se proarrhythmia. Hypocalcemia could be. Hypocalcemia prolongs the QT interval. She was not hypercalcemic. Hypercalcemia does not prolong the QT interval.
Q. Any other ways in which too much calcium can be toxic?
A. Not to the heart.
Q. What about overall?
A. I'd have -- that's internal medicine. We're going back a way.
Q. Not your thing?
A. Not my thing. But certainly, you know, I mean, acute hypercalcemia, no, but she wasn't acutely hypercalcemic, or at least I

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don't recall any serum blood tests that would have suggested that.
Q. That's not my question. We can get through this faster if you just answer the question.
A. Okay. I don't think -- I'm not sure that she -- that taking a whole bunch of Tums contributed to her death.
Q. That's not what I asked you.
A. What did you ask me?
Q. I understand your desire to advocate in every answer, but sometimes we'll just move through this faster if you just answer my question.
A. Okay. Sure.
Q. Okay. You don't have to win the case in every answer.
The question is: Is there any other way that calcium can be toxic?
I think your answer is: I don't know.
A. I don't -- it's -- acutely, I don't recall.
Q. Okay. So are you aware of the mother's statement to the medical responders

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that Ms. Moller was experiencing food poisoning?
A. I don't recall. I never saw any of the mother's statements.
Q. Okay. So you've never heard this thing about food poisoning before today?
A. No. She was -- no. I don't recall -- I don't remember that I ever heard anything about food poisoning. And I certainly never saw anything -- or read anything about -- about her mother's statements.
Q. Did you hear anything that she had been chronically vomiting?
A. Chronically vomiting, no. She was taking Phenergan. Phenergan is used as an antiemetic, so I can assume that she was nauseous. But whether or not she was vomiting or for how long, I couldn't tell you.
Q. Just because you have never heard any of that evidence?
A. I haven't heard any of that evidence. I just know what Phenergan is used for.
Q. Okay. So you're just assuming because she consumed Phenergan she must have been nauseated.

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Is that fair?
A. That's the most common reason it would be given.
Q. Yeah. What about the Benadryl?
A. Benadryl can be used for -- to help people sleep. I mean, people use -- Benadryl is over the counter. So people can use it for allergies. They can use it as a nonprescription way to go to sleep if they have insomnia because it's very soporific.
But, you know, it's over the counter. Anyone can take it.
Q. Sure. I understand that it's over the counter. That's not my question.
My question is: Is it sometimes people will take it because they're nauseated?
A. Not something I would use for nausea. It's not something I associate with people taking over the counter for nausea.
Q. What about Compazine?
A. Yeah, sure.
Q. Compazine prolongs the QT interval, correct?
A. Probably does. I'd have to look it up. Phenergan definitely does.

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Q. Okay. What about Benadryl?
A. Yes.
Q. What about Zofran?
A. No.
Q. Okay. So just -- so I made a really bad record there.
So Benadryl is known to prolong the QT interval, correct?
A. Yes, it can.
Q. Okay. And Zofran is not, according to you?
A. I have no recollection that it does.
Q. Okay. And Compazine we know does prolong the QT interval?
A. Okay.
Q. Oh, you don't remember --
A. My recollection is that it probably does. But, again, that's an easy thing to look up anytime I need to.
Q. Okay. But you haven't for this case?
A. No.
Q. Okay.
A. I could do it right now, if you'd like.
Q. I don't want you to look right now

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because I want us to just get through this.
So food poisoning.
Can food poisoning have any effect on the heart?
A. Food poisoning is speculative. I don't know what she had.
Q. Okay. That's not my question, whether you know. Listen to the questions.
A. Okay.
Q. Can food poisoning have an effect on the heart?
A. Not as a rule, no.
Q. Okay. What about an electrolyte imbalance as a result of food poisoning?
A. Yes.
Q. And, in fact, the danger of food poisoning is people get an electrolyte imbalance, correct?
A. Can, yes.
Q. And severe dehydration can be a result, correct?
A. Yes.
Q. Can severe dehydration affect the heart?
A. Yes.

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Q. Do you know what Harmony Moller's physical state was in the days and hours leading up to her death?
A. No.
Q. In your report, you say, and I quote [as read]:
"She has no recent medical illnesses and no chronic medical conditions, as was confirmed at autopsy."
Do you remember that statement?
A. I remember writing that, yes.
Q. Okay. What is your basis for that statement?
A. So when I wrote that statement, I don't believe I had seen the complete autopsy from the heart because I think there was a first initial autopsy and then a subsequent one from the organ donor people.
So when I saw at the time, I didn't see anything that in any way raised an eyebrow.
Q. What about after?
A. Afterwards, I saw the pathology from the -- from the organ donation people, and

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there was some cardiac problems, but I cannot say whether or not those were congenital and/or acquired. I have no idea.
Q. Okay. So this goes back to the issue that we got in a little of a kerfuffle over, was whether or not, in the organ donor pathology report, you can say whether this was congenital or acquired; is that right?
A. The pathologist did not opine on it. Therefore, I can be confident that this was not congenital. A congenital abnormality is something that a pathologist would report.
Also the presence of eosinophilia is not a characteristic of a congenital abnormality.
Q. So it is your opinion within a reasonable degree of medical certainty that the abnormality in Ms. Moller's heart was not congenital?
A. Yes.
Q. And it is your opinion within a reasonable degree of medical certainty that the abnormality in Ms. Moller's heart was acquired?
A. By default, yes.
Q. Are you aware of statements from

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Ms. Moller's mother that she was considering taking her to get her an IV?
A. No. As mentioned, I have not heard or read any statements by the patient's mother.
Q. Who gave Ms. Moller the Phenergan? Do you know?
A. No.
Q. Do you know who gave her the Compazine?
A. No.
Q. Same question with respect to the Zofran.
Do you know who gave her Zofran?
A. No.
Q. Do you know whether the toxicology lab screened for Zofran?
A. I don't know that they did or didn't.
Q. Do you know whether, in the autopsy, Ms. Moller's electrolytes levels were normal?
A. I don't recall any significant abnormalities. I think the potassium might have been a tad low, but certainly not dangerously so.
Q. Do you agree that potassium levels will increase postmortem?

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A. Yes, they can.
Q. What is the mechanism for that?
A. Cell death and release of potassium from cells.
Q. So just to make this clear to a layperson like me, that would mean the potassium levels will go up as time increases postmortem?
A. I would think so.
Q. Do you know what specific drugs -- illicit drugs the toxicology screen tested for?
A. I remember THC.
Q. And there was THC present, correct?
A. 9-THC I believe was there, yes.
Q. Does that have any effect on your opinion?
A. It can raise the QT interval. I know that.
Q. Anything else?
A. Other than for the fact that she -- that there was some cannabis in her system -- well, not necessarily. I think you can get that separately. No, I don't think there's anything else.
Q. Okay. Just the increase in the

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QT interval?
A. It can, yes.
Q. Do you know whether they tested for caffeine?
A. I don't know if they did, but it would not change my opinion.
Q. Okay. I'm not asking if it would change your opinion. But, I mean, I can ask that if you want me to. Sounds like you do.
A. Doesn't matter.
Q. Okay. Does the presence or lack thereof of caffeine change your opinion?
A. No.
Q. Okay. I read your opinion that more likely than not Harmony Moller died from a ventricular arrhythmia induced by consumption of kratom, especially in combination with two other agents known to prolong the QT interval; is that correct?
A. It's actually three other agents. I forgot the THC in my report.
Q. Okay. So we should add that in there?
A. Yes.
Q. So you agree with me that the

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promethazine and the THC and the other agent contributed --
A. Benadryl.
Q. -- yeah -- contributed to her death; is that right?
A. It may have, yes.
Q. And your opinion is the kratom also contributed to the QT interval?
A. Yes.
Q. And you just didn't consider whether Zofran also played a role?
A. So I don't recall ondansetron, which is Zofran, in the system. I don't recall ondansetron, in my training, as an agent on the radar for QT prolongation. If it does, we can add it. If it doesn't, it's not relevant.
Q. Okay. In your practice, do you have to order an EKG before prescribing ondansetron?
A. I don't prescribe ondansetron.
Q. Ever?
A. No. I'm not a cardiologist. Ondansetron is not a drug which is in the scope of my practice. However, anytime I look at a drug and I'm not sure, I look it up.
Q. Okay. So you know that -- in your

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report that she had a -- there was a finding she had an abnormally high left coronary ostial take-off, correct?
A. Yes.
Q. And your testimony, that's not congenital?
A. That's congenital. It's just irrelevant.
Q. Okay. But we know that there's something congenital about her heart that is abnormal?
A. No. We know that there's something congenital about her left coronary artery that is abnormal.
Q. You said something earlier, that if you came across a drug you don't really know in your practice, you would look it up.
A. Of course. I do it all the time.
Q. And that's what a responsible physician would do, correct?
A. Of course.
Q. You say that there was no anatomic cause of her death; is that right?
A. Correct.
Q. Okay. Can you help me understand

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better what that means?
A. In other words, there's nothing from the pathology report that I saw that would say this is why she died.
Q. Okay. Even though the -- I think I understand your opinion, which is even though she has a high left coronary ostial take-off, that is not related to her cause of death?
A. That's exactly what I'm saying.
Q. Now, the coroner, in the coroner's report, states that the cause of death is an abnormally high left coronary ostial take-off exacerbated by mitragynine use, correct?
A. He does say that.
Q. And that's incorrect?
A. Yeah, that's incorrect. The coronary ostial take-off has no bearing on this case.
Q. Okay. So just a mistake by the coroner?
A. It's not a mistake that the coronary might be abnormal in where it's located. It's a mistake in inferring that its presence led to death.
Q. Okay. What about the coronary ostial flap?

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A. Not relevant. Yep.
I'm sorry. I thought you finished the question. My apologies.
Q. That's okay.
I understand your opinion is it's not relevant to the cause of death. So I'm going to ask a series of questions -- which I understand that. Okay.
So does it have any significance for the person who has that condition?
A. No.
Q. So you -- you would say that you've never seen a high coronary ostia cause death.
Is that fair?
A. In itself, no, I have not.
Q. Is this possible because you see living patients and the first sign of a high coronary ostia causing death is sudden death?
A. That's not why, no.
Q. Okay. Why not? What's wrong about that?
A. Because we see patients who have unexplained cardio -- sudden death all the time and they're resuscitated. And when they are resuscitated and you evaluate them, you find

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many things.
In the world of coronary anomalies, you typically find some that are very, very well known to cause sudden death, some that rarely do, and some that you never come across.
In my entire clinical experience, I have never come across a patient or heard about a patient -- and I've been an academic cardiologist for a long time, going to morning reports with fellows on a weekly basis and hearing what comes in.
There has never been a case that has crossed my path, nor have I ever heard of anybody else's, where somebody was resuscitated from sudden death and they found an isolated coronary that was slightly high that did not travel through an intramural course, that did not travel through an interarterial course, and, most importantly, did not occur during exertion.
So, no, I think this is completely irrelevant.
Q. If the coronary abnormality that Ms. Moller had had been discovered prior to her

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death, would it have been treated?
A. No.
Q. Would it have been a reason to do a bypass?
A. No.
Q. Is there any literature you can point me to that I can read to support that this coronary abnormality is benign in Ms. Moller?
A. So with the understanding that I don't believe it's my job to give you a set of references to read, you can do your own independent research, and you will discover that in the world of coronary abnormalities and causes of sudden cardiac death, you will not find this finding.
Q. Okay. So your testimony, it's not your job to do that?
A. I don't believe it's my job to tell you which references to read. You know, I can be corrected. Let me know. But I don't believe I'm being paid to do that.
Q. Would a high ostia with a flap result in acute angle take-off with an obstruction by the flap?
A. Extraordinarily uncommon -- let me

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she has.
Q. But you would agree with me the heart was dissected and fragmented?
A. That's not relevant. The aorta was not dissected. We know where her artery came from. We know where it didn't come from. It didn't come from the right coronary ostia.
Furthermore, if she had a coronary abnormality that predisposed her to danger, it would have probably manifested during her periods as an avid soccer player, not when she's lying in bed. Coronary abnormalities don't cause sudden death at rest.
Q. Was she laying in bed when she passed?
A. Whether she was lying in bed or not, coronary abnormalities cause problems during exertion.
Q. My question was: Was she laying in bed or not when she died?
A. I believe she was. I don't know. But she wasn't playing soccer.
Q. Where was she found?
A. I believe in her room.
Q. Where in the room?

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A. I don't know.
Q. Okay. So is it your testimony that the dissection of the heart does not affect your ability to understand the course of the coronary artery?
A. That's correct.
Q. Can the presence of caffeine, Zofran, promethazine, or Benadryl have the effect of exertion on the heart?
A. I don't think I can answer the question the way you asked it.
I'm sorry. I can't answer that question.
Q. Does caffeine stimulate the heart?
A. Yes.
Q. And does Zofran stimulate the heart?
A. No.
Q. Does promethazine stimulate the heart?
A. No.
Q. Does Benadryl stimulate the heart?
A. No.
Q. Do any of these things have an effect on the functioning of the heart?
A. They can prolong the QT interval.

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They don't -- caffeine can have a modest effect on the heart rate. Other than that, no.
Q. Do you agree with me that evidence beyond what was on the labs at autopsy should be considered in trying to determine the cause of Ms. Moller's death?
A. I would think any evidence is relevant.
Q. I want to talk a little bit about the literature that you are citing for your conclusion that, more likely than not, Harmony Moller died from cardiorespiratory adverse event induced by consumption of kratom.
In your report, you state that the data and research into kratom is evolving.
Did I read that correctly?
A. Yes.
Q. And that the -- but the literature suggests a significant linkage between the main alkaloid in Mitragyna speciosa with ventricular arrhythmias?
A. There is a growing body of evidence that suggests that, yes.
Q. So -- and I think the word -- a

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growing body of evidence suggesting this, correct?
A. Yes.
Q. And you used the term "significant linkage," right?
A. Yes.
Q. Are there any meta-analyses that you're relying on for this?
A. There are what I quoted in the journal reports. There are reviews of databases from the United States and I believe Canada that look at patients who come in with kratom toxicity, sometimes as an isolated substance, sometimes in the setting of multiple substances, and what seems to be a disproportionate number of deaths, yes.
Q. And so when you're saying that, you're relying on reports to databases or poison control centers; is that right?
A. These are -- I'm just going to reference my articles here.
Q. Uh-hum.
A. Again, these are things that have been disclosed to you.
So I would look at the report,

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ATTORNEY PAYTON: Okay. I apologize for that because if I believed that, I would instruct them to stop the behavior.

THE WITNESS: I was just giving you what I had. I didn't --

ATTORNEY PAYTON: I'm not asking you what you're doing.

ATTORNEY WAGAR: It's okay. Just let's go to the next question. Okay?

- - -

(Whereupon, Exhibit 11 was marked for identification.)

- - -

BY ATTORNEY PAYTON:

Q. Marked as Exhibit 11 is a pilot placebo controlled dose finding pharmacodynamic and pharmacokinetic study from FDA.

Have you seen this before?

A. No.

Q. Okay. So fair to say you're not familiar with its content?

A. Correct.

Q. Okay. And you can see on the study that it is a dose escalation study, correct?

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A. I take your word for it.

Q. Okay. And at the dosage tested, there were no serious adverse events, correct?

A. In whom were they testing it?

Q. The testing criteria included people who had utilized some form of opioid in the past, correct?

A. You're asking me to look at this on the fly and tell you if it's correct or not?

Q. Okay. It's okay for you to say "I don't know."

A. Right. It says -- all I can see is that there were 40 subjects, and I don't know anything about their backgrounds. And it's only 40 people.

I don't know anything about their medical histories at all. I don't know if they have any congenital abnormalities. I don't know if they have any acquired abnormalities. You're just showing me a pilot study of 40 people.

Q. Okay. And you have not seen it before?

A. No.

Q. And the FDA stated that overall the

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data suggested that at the doses tested using the specific botanical kratom source for the study, under careful controlled clinical conditions, otherwise known as an inpatient study, kratom was well tolerated, correct?

A. In 40 people.

Q. Is it -- yes or no? Is that what it said?

A. Yes. In --

Q. Okay. I know you have lots of caveats that you want to put on there. I'll let your lawyer spend as much time as he wants.

A. Uh-hum.

Q. All right?

A. Yes.

Q. Do you believe -- I mean -- and the high dose is 12 grams, correct?

A. I have to look at this.

Q. Okay. Will you accept my representation to that effect?

A. I believe that you would be telling me something to be true. I obviously haven't had time to digest this and verify it, but I believe you.

Q. Okay. Do you believe that FDA would

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provide kratom to participants at such a high dose if it was concerned with significant adverse events?

A. In a healthy population, I believe that they would do that.

Q. Okay. And do you believe that FDA would provide kratom at such a high dose level if they were concerned about cardiac events?

A. So, first, it's in a population that I assume they believe it's safe to give. Humans are not a homogeneous lot. And, number two, if they're doing a study, it's to prove or disprove. Not because they know the answer a priori.

Q. Okay. I'm not sure I understood your answer. But they definitely fed the kratom to the people, right?

A. They fed kratom to 40 people who probably were -- again, I haven't looked at this -- who were probably otherwise perfectly healthy.

Q. You're saying -- you're speculating?

A. Yes, I'm speculating.

Q. You're willing to testify under oath about a study you've never even read?

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A. When the Supreme Court hears arguments, they get several months to digest it and render opinion. And you're asking me to look at something on the fly and decide how I'm going to testify. That's not fair.

ATTORNEY WAGAR: All right. Wait. I'm going to object to the form of the question. You are the one that's asking him about a study he's said several times he's never seen, and then now you're arguing with him whether he's speculating. I mean, what do you expect? You know?

ATTORNEY PAYTON: I was just surprised that he told me that the participants were perfectively healthy.

THE WITNESS: No, I said that --

ATTORNEY PAYTON: That's what I wondered --

THE WITNESS: Excuse me.

ATTORNEY WAGAR: I mean, I think he's skimming --

THE WITNESS: I said I assumed that they were perfectly healthy because, when you do a pilot study, you don't study

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people who are sick. You study healthy patients first.

I've done clinical trials. When you do a phase 1 trial and an -- early phase 2 trials, you study things on patients that are perfectly healthy before you start giving them to patients in whom you might anticipate adverse effects.

BY ATTORNEY PAYTON:

Q. You're speculating that you know what the inclusion criteria was?

A. How can I know otherwise? I've testified I've never seen this before.

Q. Okay. So you're speculating?

A. Yes, I'm speculating.

Q. In fact, in a lot of your opinions here today, you're speculating?

A. We have no hard data on a drug -- or an agent of this nature. Okay?

We do know some things about it. And it's evolving continuously. But this is not something that I can give you hard data on because it's not an FDA-approved substance. If I have -- if I have a patient who is on dofetilide, if they ask me, "Can I take

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Levaquin because I have a toothache?" I have to tell them no.

That's not speculative. I understand the drugs. I know the drug-drug interactions. These have been approved. And I can say that. Does that mean that I can't give Levaquin to 40 healthy people and say they're fine? Well, of course I can. Humans are not a heterogeneous species. I've done phase 2 trials, early phase 2 -- hear me out.

I've done phase 2 trials on patients who are perfectly healthy and have no adverse effects, no proarrhythmia. And then when I give them to patients with hypertrophic cardiomyopathy, they develop atrial fibrillation.

So you can't say I've given something to healthy people and then say, there, it's okay. Nor would you start a trial and a pilot trial by giving it to people who had comorbidities. You would want to make sure it's safe in healthy people first.

So those are my speculations based on my experience of having been a doctor for as long as I have been. So, yes, they are

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speculations, but they are reasonable speculations, Counselor.

Q. Okay. And your opinion is based on your reasonable speculations?

A. Correct.

Q. And your reasonable speculations that are formed by your experience, right?

A. Yes.

ATTORNEY WAGAR: Let me get a word in edgewise here. I'm going to object to the form. You're not specifying which opinions. You're just, you know, generally --

ATTORNEY PAYTON: Okay. Let me make it more clear.

BY ATTORNEY PAYTON:

Q. I'm talking about your opinions here in this case today.

ATTORNEY WAGAR: All of them. All of them?

BY ATTORNEY PAYTON:

Q. Go ahead and answer.

A. What's the question?

Q. When you're talking about you're relying on speculation that is informed by your

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experience as a medical doctor and your experience and expertise in the field, your opinions here today are based on speculation.

And your opinions about the cause of death of Ms. Harmony Moller are based on speculation being --

A. Plus the literature which I've cited, which is published not only in journals but very high-impact journals. The Journal of the American College of Cardiology is the premier cardiology journal of the United States. This is not a throwaway journal. This is peer-reviewed, valid stuff with good editorials.

Q. Okay. Let me try my question again because I didn't ask it really super well.

Your opinions here today about the cause of death of Ms. Harmony Moller are based on speculation informed by literature and your experience as a cardiologist, correct?

A. Yes.

Q. Let's go back to the Krantz article, which is one of the articles that you relied on and you just told me about in the two retrospective database analyses, Li and Krantz,

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correct?

A. Sure, uh-huh.

Q. Yes. And will you agree with me that as of February 2023, when Ms. Moller died, no regulatory agency had issued any warnings regarding kratom potentially causing arrhythmia?

A. That's probably true. I don't know that it did or didn't. Probably not.

Q. In fact -- oh, sorry.

A. I have no reason to disbelieve you.

Q. Okay. And, in fact, the Krantz article states, and I quote [as read]:

"To date, no regulatory agency has issued warnings regarding proarrhythmic properties of kratom despite scattered reports of unexplained sudden cardiac death."

Correct?

A. Uh-hum.

Q. Is that a "yes"?

A. Yes.

Q. Okay. The study, I think you already

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told me, relied on publicly available data through the FDA Adverse Event Reporting System, sometimes called FAERS, F-A-E-R-S, the Canadian Vigilant [sic] Adverse Reaction online database, and the Center for Food Safety and Applied Nutrition Adverse Event Reporting System, correct?

A. Yes. That's in the abstract of the article.

Q. And these are databases where people can call in and say something happened, correct?

A. Yes. I believe a lot of those reports come from emergency departments as well.

Q. But they don't verify any medical records along with these reports, correct?

A. The reports have to do with sudden death. They verified the arrythmias.

Q. So your testimony is the databases will ask for medical records before they report?

A. No. My testimony is, in the Krantz article, that there was disproportionate amount of ventricular arrhythmias.

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Q. I'm asking you a different question.

How did these reports come to be? It's because people call in and make a report, correct?

A. I assume so, yes.

Q. They don't -- at the reporting database, they note everything down that is called in, right?

A. Yes.

Q. They don't ask for medical records, correct?

A. No.

Q. And people individually can call in and report, correct?

A. These are reports that are coming in large from -- they're not -- they're being called in from emergency rooms. These are being called in from -- after people have sudden death.

Q. Would you be surprised that the vast majority of calls to all of these databases come from individual consumers?

A. A little surprised.

Q. Okay. And do you know --

A. I mean, consumers aren't in the

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cranberry hasn't been showing up in the cardiology literature as a reason for concern. Kratom has been.

Q. What about cannabis? Has cannabis been approved?

A. Regrettably, in several states, yes.

Q. I'm not asking about legality but approved to treat a medical condition.

A. No.

Q. Okay. And cannabis certainly shows up in medical situations, correct?

A. Sure.

Q. Has ayahuasca been approved to treat any medical conditions?

A. I've never even heard of it.

Q. How about psilocybin? Any approval to treat --

A. Who?

Q. Psilocybin.

A. I've never even heard of that one.

THE COURT REPORTER: Can we take a break?

ATTORNEY PAYTON: Yeah, for sure. How long do you want?

THE COURT REPORTER: Five

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minutes is fine.

THE VIDEOGRAPHER: The time right now is 1:38 p.m. We're off the record.

- - -

(Whereupon, a short recess was taken.)

- - -

THE VIDEOGRAPHER: The time right now is 1:47 p.m. We're back on the record.

BY ATTORNEY PAYTON:

Q. Of the 46 reports in the Krantz article, do you know if any involved underlying disease?

A. No.

Q. And you agree that none of these reports have been verified, correct?

ATTORNEY WAGAR: Object to the form. What do you mean "verified"?

BY ATTORNEY PAYTON:

Q. Did the study -- the database go back and verify with medical records or autopsy that the reports were, in fact, correct?

A. Well, you can't verify ventricular

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fibrillation on an autopsy. You can verify it on an EKG, but not an autopsy.

Q. I'm talking about even broader.

They don't go back and take a look and make sure that the report itself was accurate?

A. Probably not.

Q. Okay. There's no real process to verify these reports, correct?

A. Not that I'm aware of.

Q. You talk about QT prolongation, correct, in your --

A. Yes.

Q. And, in fact, the Krantz study shows a complete lack of association of mitragynine to QT prolongation, correct?

A. Yes, but this patient, Ms. Moller, had three other substances that can synergistically affect the QT interval. Therefore, one has to consider that.

Q. Okay. But that really wasn't my question. My question was really simple and straightforward.

There is no association whatsoever in the Krantz report between

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mitragynine and QT prolongation?

A. No. The article was primarily primary ventricular fibrillation.

Q. Okay. There was one report of mitragynine-related QT prolongation, correct?

A. One that I shared. There may be others, yes.

Q. No. I'm talking about in the Krantz report.

A. Oh, in the Krantz report. Yes. Yes.

Q. Okay. One out of 5 million?

A. Uh-hum, yes.

Q. Okay. So -- and the Krantz report relied on a study that found, and I quote [as read]:

"No evidence of an increased incidence of QT prolongation."

Right?

A. Yes.

Q. And that's the Abdullah, A-D-B-U-L-L-A-H [sic], report that you cited to me earlier. Am I right?

A. Yes. I read that report.

Q. Okay. And you did read that report,

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death, not a sparsity recording to sudden cardiac death from ventricular arrhythmias.
Q. So -- but then look with me on page 2266, when they have the discussion of the study limitations to interpret being arrhythmia risk signals.
Do you see that?
A. Yes, of course.
Q. It states that [as read]:
"Pharmacovigilance data, although providing large numbers of reports, cannot measure incident, and causal determinations cannot be made."
Correct?
A. Yes, it does say that. However --
Q. I'm just asking. Does it say it or not?
A. Yes, it does, but does that mean I -- however, the same paragraph --
Q. You get a chance to have your lawyer ask the questions that he wants to ask. I understand you want to say all of these things. And if you want, I can give you an open-ended

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question where you can advocate and do whatever you want, get it out of your system. But I really want to run through my questions.
A. Okay.
Q. Is that fair?
A. Yes.
Q. Okay. And then -- so we know that this study doesn't purport to suggest anything on causal determinations, correct?
A. Can you ask that question again?
Q. We know that the Krantz study does not purport to suggest anything on causal determinations?
A. I'm not sure I can agree or disagree with that.
Q. Well, I'm quoting their own words. [as read]:
"Causal determinations cannot be made."
Correct?
A. It says causal determinations cannot be made earlier in that paragraph, yes.
Q. They also note that additional risky behaviors, like unmeasured illicit drug ingested, can't be caught in this database,

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correct?
A. Correct.
Q. And it's all based on self-reporting?
A. Yes.
Q. With no verification?
A. You shouldn't have to verify ventricular fibrillation, but yes.
Q. And association doesn't equal causation, correct?
A. Correct.
Q. When you do a FAERS data analyses, the FDA states that there is no certainty that a suspected drug caused the event, correct?
A. Where does it say that?
Q. It's in the FDA materials. I can show you if you don't believe me.
A. I believe you.
Q. Okay. And the event could, in some cases, be caused by an underlying disease or another drug, correct?
A. Not impossible.
Q. Given this from the source alone, you cannot conclude that mitragynine causes VT/VF, correct?
A. Absence of proof isn't proof of

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absence.
Q. I'm just asking -- sorry. Go ahead.
Are you really going to say that this study concludes that mitragynine causes VT/VF?
A. I think the study demonstrates that there's a need for more vigilance, because it may very well.
Q. Certainly there's association, right?
A. Yeah.
Q. Okay. But not causation?
A. It raises an eyebrow.
Q. My question is: Is it proof of causation?
A. No.
Q. The article also notes that there is a lack of a convincing mechanistic link between mitragynine and ventricular arrhythmia, correct?
A. It needs to be studied more firmly.
Q. No. My question is: The study notes that there is convincing mechanistic link -- there is a lack -- sorry -- of a -- let me try it again.
The study notes that there is a

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lack of a convincing mechanistic link between mitragynine and ventricular arrhythmia, correct?
A. It says specifically that the data may represent underappreciated triggers for malignant arrhythmias, and although a convincing mechanistic link is lacking, the pharmacovigilance findings suggest that additional clinical research is necessary.
Q. Nobody is denying that we want to do more research, correct?
A. Correct.
Q. And, in fact, are you aware that one of the biggest supporters of more research in this case are the defendants?
A. I wouldn't know that.
Q. Okay. But would it surprise you?
A. Yes.
Q. Why is that?
A. They may not like what they get.
Q. They also may like what they get, correct?
A. Correct.
Q. Okay. If you want to say there are these weak signals on cardiovascular disease,

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there's also signals on risk mitigation, correct?
A. Risk mitigation? I'm not sure I understand what signals there are in risk mitigation.
Q. Numerous people reporting that this has been an effective way they've beaten opioid addiction, correct?
A. I can't confirm or deny.
Q. Right. It's still speculative?
A. No. It's not speculative. I know that it's used as an opioid substance to treat -- not to treat. It's a substance that's used for opioid use disorder despite the fact that the FDA has three substances that are actually FDA-approved for that purpose.
The difference is that if there are small -- and they may be very small -- populations that are vulnerable to effects of these agents because they have not fully been studied, we don't know what those populations are.
If there are people out there who have a congenital abnormality or an acquired abnormality or a channelopathy, such

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as like a baseline prolonged QT interval or something like that, those are the populations in whom we don't know that it's safe or not. When you quote to me the 5 million here or there, I have no reason to disbelieve you.
The difference is if there's a small but real population out there for whom this may not be appropriate, we don't know who these people are, just as I can take a dose of Levaquin, but some of my patients can't.
That's all.
Q. In fact, is Suboxone safe for all people?
A. I'm sorry?
Q. Is Suboxone safe for all people?
A. Is who?
Q. Suboxone.
A. Suboxone? No.
Q. Is Buprenorphine safe for all people?
A. These are drugs that are approved. They have guidelines.
Q. No. I'm asking you a different question. I didn't ask if they were approved. I don't deny that.
A. Right. But that's not the point.

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Q. It's my question. It's not what you want to say. It's my question.
Is Buprenorphine safe for all people? Yes or no?
A. Can I look it up?
Q. It's okay if you say "I don't know."
A. I don't know. But I can look it up? And if there are indications and contraindications, I can find that out because it's already been adjudicated. So I know -- so I can look that data up here because I know -- that's been looked at.
Q. Have you been on your phone during this deposition?
A. Never.
Q. Okay. Just cause you're waving it around.
A. Because I'm saying that's how I look things up. You know, I have a drug database on my phone. Most doctors do.
So with a medication that's been approved, I know in whom something is safe and in whom it's not. Just like my simple examples with an innocuous antibiotic like Levaquin.
Q. Your testimony is for Suboxone, all

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A. Let's see. Yes, 30 minutes after his last dose of kratom, he experienced approximately two minutes of witnessed seizure-like activity. That's right.
Q. And he self-reported ingesting kratom daily for the past four years, right?
A. Yes.
Q. And he reported a daily intake of approximately 84 grams to 100 grams, correct?
A. Yes.
Q. How does that compare to the 52 nanograms per milliliter in Harmony Moller's blood?
A. It's much more.
Q. Significantly more, right?
A. Yes.
Q. Okay.
A. But, again, Ms. Moller did have three other substances that can synergistically prolonge the QT interval. I know that's not your question, but I think --
Q. I know you love to say that.
A. -- but I think it's important to make that additional reference.
Q. Do you feel comfortable that you've

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made that point now?
A. Yes.
Q. Good. So then we can move along and try to get through this.
Now, his -- this patient's initial electrocardiogram revealed a Type 1 Brugada pattern, correct?
A. Yes, with a prolonged QT.
Q. Okay. So we have one patient who took a significantly higher amount of kratom than Ms. Moller, correct?
A. Yes.
Q. And the article references three other meeting abstracts reporting kratom use masking a Brugada pattern, correct?
A. Yes.
Q. But it notes that no testing was done to rule out contaminants, right?
A. Yes.
Q. And are three meeting abstracts sufficient evidence for you, considering the estimated 5 million users?
A. Again, it just points to the fact that there may be a very small but real susceptible population.

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Q. You also pointed -- there's an editorial on this case report too, correct?
A. Yes.
Q. Published when? October 2024?
A. The editorials are always typically written in the same -- the same -- the same volume -- the same volume. So no -- published at the same time, March -- I believe it was the same time. Let me see.
Yeah. March. March 2025.
Editorial comments usually happen in the exact same article, not months later.
Q. And this article makes your point, which is more studies are needed, right?
A. Yes.
Q. Okay. And are you aware of any published prospective clinical pharmacokinetic studies --
A. Pharmacokinetic.
Q. -- that associates the use of kratom products with ventricular tachycardia?
A. Prospectively?
Q. Correct.
A. No. No prospective studies, I believe, of that nature have been done.

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Q. Any other studies you want to point me to?
A. Not right now.
Q. Okay. Are you aware of any?
A. I'd have to look.
Q. But you're here for your deposition today opining on your opinion, right?
A. Yes.
Q. And you're prepared?
A. Yes. Tired, but prepared.
Q. Okay. So let me ask you about ones that I think might be possibly relevant, and you tell me if you're familiar.
Are you aware of a 2024 study "Human mitragynine and 7-hydroxymitragynine pharmacokinetic after single and multiple daily dose of oil-encapsulated dried kratom leaf powder"?
You told me about it earlier today.
A. I don't remember. I don't remember.
Q. That was the Huestis study. I know it was kind of --
A. I'm sorry. Is that the Huestis study that we started with?

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large amounts of patients with different comorbidities, that we have a patient who had sudden death with no real plausible other explanation, based on my experience as a cardiologist.

Don't forget. Ms. Moller was not my patient. That's all I am saying, that there is a signal in the cardiology literature that is growing that suggests that there is a disproportionate amount of sudden cardiac death relative to the amount of kratom that has been out there that is coming -- that is coming out, that is emerging in the cardiology literature.

That's what I'm saying.

Q. That summarizes your opinion right there?

A. I think so, yeah.

Q. Okay. So you cannot say within a reasonable degree of medical certainty that in this case kratom caused Ms. Moller's death, correct?

A. I have every reason to believe that it did because of lack of anything else on the pathology that would have done so.

Again, people don't die for no

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reason. She had no anatomic reason for sudden death. They didn't have a bleed in her brain. She didn't have coronary artery disease. She didn't have a heart attack. She didn't have a stroke. She didn't have trauma. She didn't have a chronic metabolic process like cirrhosis or end-stage renal disease.

She died from something. And arrhythmias are the perfect crime because they don't leave a trace.

She had a drug that can cause primary ventricular arrhythmias, which is still very, very possible. The Krantz report is -- demonstrates, yes, no perfect cause, but clearly an association between primary ventricular arrhythmias.

In addition, we know that it can prolong the QT interval. In most patients, it does not. But she had three other substances on board which can act synergistically.

Q. But my question is a little bit more specific. I hear you that there is a signal in the literature that is associating kratom use with cardiac event.

We agree on that, correct?

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A. Yes.

Q. Okay. But that's different than saying that you, as a medical doctor sitting here today, can say within a degree of reasonable medical certainty that Ms. Moller died from kratom. That's an additional --

A. I can't --

Q. Can you say --

A. Right.

But I can't think of anything else.

Q. Aside from not thinking of anything else, can you testify to a degree of medical certainty that it was the kratom?

A. In combination with the other drugs that she had in her system, I think so, yes.

Q. And the other drugs -- at least some of the other drugs in her system are well-known to increase the QT interval?

A. Yes.

Q. Here we have something that is not well-known to increase QT interval?

A. It can. And it doesn't have to a lot when you have three other substances on board that do the same thing.

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Q. We just went through all your literature that you point to, and I didn't see a single thing that said kratom increases the QT interval.

What am I missing?

A. We just did. The last article. Granted, the levels were much higher, but it did.

Q. The one case report?

A. Yeah, we just did. We went through that.

Q. Okay.

A. We know that it blocks the channel that can lead to QT prolongation. In most patients, it does not. In some patients, it --

Q. You're relying on Miller for that?

A. I'm relying on Miller and also the fact that as a woman she is more likely to be susceptible to a longer QT interval. And as someone who is taking three other drugs, that can cause the prolongation of a QT interval. The addition of a fourth, even at a modest level, can also prolong the QT interval.

And, finally, we don't even know, nor will we ever, if she had a genetic

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cause for a basis of a prolonged QT interval because that's only something -- you can't find that out now. You had to find that out when she was alive. I never saw an EKG on her ever, so we don't know.

I'm saying that she may have been a vulnerable patient for sure, a minority, a big minority of the people that's out there. But, nonetheless, she may have been a vulnerable patient as well. But I can't think of anything else that would have done this by process of elimination.

Q. And that's your opinion?

A. Yes, that's my opinion.

Q. Okay. Let's talk about your rebuttal opinion that you issued.

A. Okay.

Q. I wanted to ask you a few questions about the hypertrophic cardiomyopathy.

A. Please.

Q. So does -- is it a typical manifestation of the hypertensive cardiomyopathy change in the right ventricle?

A. Did you say hypertensive or hypertrophic?

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Q. I'm sorry. I said hypertensive, and I made a mistake. Can I try again?

A. Please.

Q. Let's talk about hypertrophic cardiomyopathy.

Does that affect the right ventricle?

A. Rarely.

Q. Okay. Is it typical?

A. No.

Q. Okay. But it can?

A. Not impossible.

Q. Okay. What is the mechanism for the arrhythmia to occur?

A. In hypertrophic cardiomyopathy?

Q. Correct.

A. Hypertrophic cardiomyopathy typically leads to ventricular fibrillation when people have sudden cardiac death.

Q. Is hypertrophic cardiomyopathy uniform through the left ventricle?

A. Doesn't have to be, no.

Q. Okay. Is it possible that the septum is not the best place to biopsy?

A. The septum is usually the first place

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that it happens, actually.

Q. Yeah. But is it the best?

A. I'm not sure I understand your question.

Q. Yeah. If you were going to do a biopsy, would you think that there would be a better place to do the biopsy on than the septum?

A. No. That would probably be the number one place you would do it.

Q. And why is that?

A. Because the septum is almost always affected.

Q. Would you -- is the septum always affected?

A. Overwhelmingly majority of the time, yeah. In medicine, we never say always. But the septum is very rarely normal in hypertrophic cardiomyopathy.

Q. But it can be?

A. It depends on where in the septum too. The base -- the septum is a long wall. It goes from the base of the wall to the apex. As you get -- you know, it's very variable. As a rule, the septum is a pretty good place to

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biopsy.

Q. So even within the septum itself, you might biopsy in an affected or unaffected portion?

A. Unlikely. No. Even if you don't have hypertrophy, the hallmarks of hypertrophic cardiomyopathy are metabolic -- are cardiac -- myocyte disarray is what they call it. The biopsy was very clear that this is not myocyte disarray.

Furthermore, the mitral valve was reported as normal. Patients with hypertrophic cardiomyopathy invariably have abnormalities in their mitral valve anatomy.

The LV cavity was reported as normal. The LV cavity in hypertrophic cardiomyopathy is typically small. The incidence of sudden cardiac death in hypertrophic cardiomyopathy is still quite rare, and when it is happening, the walls are considerably thicker than what was reported.

Q. Okay. But you're answering a lot of questions that aren't the question that I asked.

But let me ask the questions.

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what she's asking and we'll get through this.
BY ATTORNEY PAYTON:
Q. Okay. So would you agree with me, if you're right on the science, you're right, correct?
A. Yeah.
Q. Because science is objective?
A. Yeah.
Q. And you can rest assured that if you are saying true science, the truth will come out, correct?
A. Yes.
Q. So you don't need to be advocating or answering other questions?
A. Correct.
Q. Okay. So let's try to stay on track here.
I want to look at something specific you said in your report that you --
A. This is the rebuttal?
Q. Yeah. The rebuttal. The supplemental, the -- no, it's a rebuttal. Sorry.
Okay. You note that there's a

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growing body of evidence linking kratom to hypertension, correct?
A. Yes.
Q. And which could lead to hypertensive heart disease, correct?
A. Yes.
Q. And that would be left ventricular hypertrophy?
A. Yes.
Q. And then you go on to say, and I quote [as read]:
"It is certainly possible that the left ventricular hypertrophy noted in the autopsy was a manifestation of kratom exposure."
Correct?
A. Only if she been using it for a long time. I have no idea how long she was using it.
Q. It's just a speculation?
A. Hypertension, whether she has hypertension because she has hypertension like a third of American adults have or if it was from kratom exposure or any combination

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thereof, can cause left ventricular hypertrophy.
Q. Okay. So you don't know one way or the other if it was caused by kratom or something else, correct?
A. No. It could just be from hypertension.
Q. Okay. Do you rely on anything other than the Chicagi article for the proposition about the growing body of evidence linking kratom to hypertension?
A. I have to look. I believe that I saw several reports that it causes hypertension, but I don't remember. I'll be honest.
Q. Okay. Are you able to tell me here in your deposition?
A. I don't remember. I'd have to look it up.
Q. But we know that there is this one article published in November of 2023, correct?
A. I don't remember when it was published.
Q. Okay. And it's in the Journal of Addictive Diseases?
A. Uh-hum.

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Q. Yes?
A. Yeah, but I did as much of a research as I could when I was looking this stuff up.
Q. So this is the best you could come up with?
A. Afraid so. It's hard to come up with lots of stuff on a substance that's not regulated.
Q. I mean, there we are again; not regulated.
What do you mean by that?
A. There are no large prospective studies.
Q. Okay. That's different than not regulated, right?
A. Sorry.
Q. Do you consider the Journal of Addictive Diseases a credible source in the medical literature?
A. Yeah. It's not -- you know, it's not a -- it's not a high-impact journal, but it's not unreliable. It's peer-reviewed.
Q. Do you regularly rely on articles from that journal in your clinical work?
A. No, because kratom is not usually

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don't treat thickening per se. You know, you could get -- the likelihood of her having -- I don't know. I mean, I just -- the answer to that is, you know, you look for a treatable cause. The most common would be hypertension. Other than that, I can't think of anything.

Q. Is there any evidence that she had ever been diagnosed with hypertension before her death?

A. I don't know that she ever saw doctors. I don't think so. I don't know.

Q. There's no evidence of hypertension in this particular case as far as it being diagnosed prior to her death?

A. No. I mean, athletes can have hypertension, have had thickening of the walls too. I don't know why her walls were moderately thick.

Q. Even assuming she had some hypertension, is that, in your opinion, likely to have had anything to do with the cause of her death?

A. Of course not.

Q. Now, you were asked before about whether your conclusions amounted to

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speculation.

I'm going to ask you to assume for a moment that, you know, in law, speculation, when it comes to giving an opinion in a case, involves a conclusion or an opinion that is not based on sufficient evidence, facts or reliable foundation. Okay?

A. Okay.

Q. Is that what you meant about your opinions in this case?

ATTORNEY PAYTON: Object to the form of the question, and object to the instruction to the witness. Object to the leading question.

BY ATTORNEY WAGAR:

Q. You can answer.

A. Of course not.

Q. Okay. Did you, in fact, use a methodology that a cardiologist would normally use in your profession to determine what her cause of death was in this case?

A. Absolutely.

Q. And did that methodology include her autopsy, the toxicology screen, and all the evidence that we've talked about today?

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A. Yes. It was conclusion based on the totality of the evidence and lack of any other logical explanation.

Q. Okay. And, therefore, you do believe more likely than not that Ms. Moller died of an arrhythmia that was caused, in whole or in part, by her ingestion of kratom with other substances co-ingested prior to her death?

A. Yes. That's exactly what I said.

Q. Okay. Are you familiar with -- as a medical doctor with the term "differential diagnosis"?

A. Of course.

Q. Could you just briefly explain what a differential diagnosis is and how that methodology informed your opinion in this case.

A. So a differential diagnosis is simply a list of possible explanations for a clinical picture.

Q. And did you use that methodology in this particular case?

A. Yes.

Q. And were you able to eliminate any other cause other than the one you feel more likely than not caused her death?

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A. That's exactly the point. I was not.

Q. Okay. And differential diagnosis is a -- is a methodology that's, I think, taught in medical school, isn't it?

A. Yes, it is, very early on.

Q. It's fundamental to medicine when it comes to diagnosing conditions that are not obvious, like a gunshot to the head or a broken arm, right?

A. Correct.

ATTORNEY PAYTON: Objection. Leading.

BY ATTORNEY WAGAR:

Q. And so the methodology used, as far as coming to a conclusion in this particular case, is based upon a methodology using differential diagnosis and eliminating any other possible cause?

ATTORNEY PAYTON: Objection. Leading.

THE WITNESS: Correct.

BY ATTORNEY WAGAR:

Q. Is that correct?

A. Yes. That's absolutely correct.

Q. Okay. So it would be incorrect,